Lab 10

Objective: Learn about arrays of pointers to strings.
This lab builds on lectures 8, 9, and 10

Exercise 1: Finding Longer-Than-Average Words
You should write a program that:

1. Reads, from `stdin`, up-to 100 whitespace separated words and stores them in an array of `char *`. Yes, that means you'll need to call `malloc()` in order to allocate each string. No word will be longer than 79 characters. All input will be in ASCII. Use Google (or your favourite search engine, or ChatGPT) to find out how to read strings using `scanf()` in C.

2. Calculates the average length of a word in that input set.

3. Prints to `stdout` the average length of a word (to 2 decimal places). Use Google (or your favourite search engine, or ChatGPT) to find a web page that tells you which `printf()` format string you should use.

4. Prints to `stdout` each word that is longer than the average length. These should be output in the same order they were input.

5. Deallocates all the space it used.

6. Returns cleanly to the operating system.

If no words are entered, then there should be no output at all.